

CONNECTIVE TISSUES

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Connective Tissues

1-Introduction

Connective tissues are ubiquitous tissues, the most widespread in the body; they are composed of a mixture, in variable proportions, of cells, fibers, and ground substance.

They are of mesenchymal origin.

Connective tissues perform several functions, including:

Connection between organs: example, tendons.

Packaging and support: example, capsules and framework.

Nutrition: they constitute the pathway for interstitial fluid.

2. Constituent Elements of Connective Tissue Proper

2.1 Ground Substance

The ground substance is homogeneous and amorphous, occupying the spaces between the cells and fibers of connective tissue. It is a filling material secreted by fibroblasts. It contains substances such as collagen and elastin, proteoglycans and glycoproteins, polypeptides, amino acids, glucose, water, mineral salts, and exogenous substances derived from blood plasma.

Proteoglycans are polypeptide chains onto which complex molecules of glycosaminoglycans are attached.

2.1.1 Glycosaminoglycans (GAGs)

Six types are described (sulfated and non-sulfated):

Hyaluronic acid

Chondroitin sulfates

Dermatan sulfates

Heparan sulfate

Keratan sulfates

Heparin

2.1.2 Proteoglycans

2.1.3 Glycoproteins

The best characterized is fibronectin, an adhesion molecule with binding sites for cells and fibers, linking them to the ground substance; laminin, as well as fibrous proteins such as procollagen and proelastin

In pathology:

The ground substance may accumulate in recessive anatomical diseases called mucopolysaccharidoses, including: Hunter disease, Hurler disease, Morquio disease

2.2 Connective Fibers

2.2.1 Collagen Fibers

Light microscopy (LM): fibers are demonstrated by eosin, saffron, and Masson's trichrome. They are thick fiber bundles, extensible but non-elastic, parallel and never anastomosed. They provide connective tissues with resistance and strength against mechanical traction forces and intermittent pressure.

Electron microscopy (EM): fibers are composed of juxtaposed fibrils, themselves formed by the assembly of microfibrils showing alternating light and dark bands with a specific periodicity.

Microfibrils result from elementary molecules: tropocollagen.

Tropocollagen is formed by the triple helix winding of three alpha polypeptide chains.

Procollagen synthesis occurs intracellularly (in fibroblasts), whereas the assembly of tropocollagen molecules occurs extracellularly (in the extracellular matrix).

Formation of Collagen Fibers

Alpha chain subunits are synthesized by fibroblasts and assemble into three α polypeptide chains forming a triple helix to produce procollagen molecules.

After exocytosis, peptidases cleave the amino and carboxyl ends of alpha chains to form tropocollagen, the fundamental unit of collagen (rich in glycine, hydroxyproline, and proline).

In the extracellular space, tropocollagen molecules assemble end-to-end and side-by-side with a 67 nm overlap, responsible for the characteristic striated periodicity of collagen.

Fibers are linked by lateral interactions, making collagen bundles extremely resistant.

Collagen resistance and flexibility depend on its physicochemical properties. Several types exist:

Type I: most abundant; thick fibers resistant to traction. Found in dermis, tendons, bone tissue, and dentin.

Type II: thin fibers resistant to intermittent forces, located in cartilage.

Type III: reticulin fibers (0.5–1 μm in diameter), found in hematopoietic organs, blood vessels, lungs, and liver.

Type IV: specific to basal laminae.

2.2.2 Reticulin Fibers

Observed under light microscopy after silver impregnation, they appear thinner than collagen fibers. Chemical analysis reveals tropocollagen molecules synthesized by fibroblasts. Reticulin fibers are fine type III collagen fibrils, isolated or grouped in small bundles, acquiring a glyco-lipid layer that prevents their transformation into collagen fibers. They are found mainly in hematopoietic organs (lymph nodes, spleen, bone marrow).

2.2.3 Elastic Fibers

Demonstrated by orcein staining, their diameter ranges from 0.2 to 2 μm . They are the thinnest fibers and less abundant than collagen fibers. They branch and anastomose extensively, forming networks. Elastic fibers consist of a central elastin region (hydrophobic protein) and a microfibrillar glycoprotein component (fibrillin). They are found in the yellow intervertebral ligament, elastic cartilage, and walls of large arteries.

2.3 Connective Cells

2.3.1 Fibroblasts and Fibrocytes

Spindle-shaped cells with branched extensions. Under electron microscopy, fibroblasts show abundant organelles involved in protein synthesis. They synthesize connective tissue matrix components, including procollagen and proelastin.

In mature connective tissue, fibroblasts become less active and are called fibrocytes, characterized by a denser nucleus and reduced organelles.

2.3.2 Adipocytes

Cells specialized in lipid storage, grouped within connective tissues. Two types exist:

White adipocytes Round or oval cells with a large lipid vacuole and a small peripheral nucleus.

Brown adipocytes Smaller cells with a central nucleus, numerous small lipid vacuoles, and abundant mitochondria. Present during fetal life to generate heat at birth.

2.3.3 Immune Cells : Loose connective tissues are major transit sites for immune cells.

2.3.3.1 Macrophages

Derived from blood monocytes; large irregular cells with eccentric kidney-shaped nuclei, abundant vacuolar system, and pseudopodia.

Functions:

Phagocytosis

Antigen presentation to T lymphocytes

Cytokine synthesis

2.3.3.2 Plasma Cells

Terminally differentiated B lymphocytes, specialized in immunoglobulin secretion. Oval cells with eccentric “cartwheel” chromatin nucleus, abundant Golgi apparatus, and rough ER.

2.3.3.3 Mast Cells

Rounded cells with central nucleus and cytoplasmic granules rich in histamine. Abundant in skin, bronchi, and digestive tract; involved in allergic diseases.

2.3.3.4 Other Cell Types Lymphocytes, Monocytes.

3. Varieties of Connective Tissues

3.1 Loose Connective Tissue

Balanced proportions of fibers, cells, and ground substance.

3.2 Dense Regular Unidirectional Connective Tissue

Parallel collagen fibers (tendons, ligaments).

3.3 Dense Regular Bidirectional Connective Tissue

Perpendicular fiber layers (corneal dermis).

3.4 Dense Irregular Connective Tissue

Thick interwoven collagen fibers (organ capsules).

3.5 Reticular Connective Tissue

Network of reticulin fibers supporting hematopoietic organs.

3.6 Elastic Connective Tissue

Predominance of elastic fibers (large arteries).

3.7 Primary Adipose Connective Tissue

Brown fat, abundant in fetus, thermoregulation.

3.8 Secondary Adipose Connective Tissue

White fat (hypodermis), metabolically active.

3.9 Decidual Connective Tissue

Forms in uterine mucosa during pregnancy.

3.10 Mucous Connective Tissue

Rich ground substance; found in Wharton's jelly of the umbilical cord.